

CLASS TEST - 2

① (a) Cauchy-Riemann conditⁿ in Cartesian coordinates :

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad ; \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{--- (i)}$$

In spherical coordinates :

$$x = r \cos \theta \quad ; \quad y = r \sin \theta \quad \text{--- (ii)}$$

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\boxed{\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta} \quad \text{--- (iv)}$$

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$\boxed{\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x} (-r \sin \theta) + \frac{\partial u}{\partial y} (r \cos \theta)} \quad \text{--- (v)}$$

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\boxed{\frac{\partial v}{\partial r} = -\frac{\partial v}{\partial y} \cos \theta + \frac{\partial v}{\partial x} \sin \theta} \quad \text{--- (vi) [from (i) & (ii)]}$$

$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial \theta} = \boxed{\frac{\partial v}{\partial y} (r \sin \theta) + \frac{\partial v}{\partial x} (r \cos \theta)} \quad \text{--- (vii)}$$

From (i) - (vii):

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} ; \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

— CR condition in spherical polar coordinates

(b) Let $z = re^{i\theta}$

$$\begin{aligned} f(z) &= z z^* \exp\left(\frac{1}{z} - \frac{1}{z^*}\right) \\ &= |z|^2 \exp\left(\frac{z - z^*}{|z|^2}\right) \\ &= r^2 \exp\left[\frac{r(e^{i\theta} - e^{-i\theta})}{r^2}\right] \end{aligned}$$

$$f(r, \theta) = r^2 \exp\left(\frac{2i \sin \theta}{r}\right)$$

CR condition in spherical coordinates:

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} ; \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$$

Here, $u(r, \theta) = r^2 \cos\left(\frac{2 \sin \theta}{r}\right)$ [expanding $\exp\left(\frac{2i \sin \theta}{r}\right) = \cos\left(\frac{2 \sin \theta}{r}\right) + i \sin\left(\frac{2 \sin \theta}{r}\right)$]

$$v(r, \theta) = r^2 \sin\left(\frac{2 \sin \theta}{r}\right)$$

$$\therefore \frac{\partial u}{\partial r} = 2r \cos\left(\frac{2 \sin \theta}{r}\right) + \cancel{r^2} \sin\left(\frac{2 \sin \theta}{r}\right) \cdot \left(-\frac{2 \sin \theta}{r^2}\right)$$

$$\frac{\partial u}{\partial r} = 2r \cos\left(\frac{2 \sin \theta}{r}\right) - 2 \sin \theta \sin\left(\frac{2 \sin \theta}{r}\right)$$

$$\therefore \frac{\partial v}{\partial \theta} = r^2 \cos\left(\frac{2 \sin \theta}{r}\right) \cdot \frac{2 \cos \theta}{r}$$

$$\frac{\partial v}{\partial \theta} = 2r \cos \theta \cos\left(\frac{2 \sin \theta}{r}\right)$$

$$\therefore \frac{\partial u}{\partial r} \neq \frac{1}{r} \frac{\partial v}{\partial \theta} \rightarrow \text{CR condition is not satisfied}$$

Hence, $f(z)$ is not an analytic func.

(2) (a) $i^i = (e^{i\pi/2})^i = e^{-\pi/2}$

$$\therefore \operatorname{Re}[i^i] = e^{-\pi/2} ; \quad \operatorname{Im}[i^i] = 0 \quad (\text{Ans})$$

(b) $\log(-1) = \log(e^{i\pi}) = i\pi$ [$n \rightarrow$ odd integer]

$$\therefore \operatorname{Re}[\log(-1)] = 0 ; \quad \operatorname{Im}[\log(-1)] = \pi \quad (\text{Ans})$$

as $\cos(n\pi) = -1$

$$\begin{aligned}
 (b) \quad z^n + \frac{1}{z^n} &= e^{in\theta} + e^{-in\theta} \\
 &= \cos(n\theta) + i \sin(n\theta) + \cos(-n\theta) + i \sin(-n\theta) \\
 &= 2 \cos(n\theta)
 \end{aligned}$$

$z^n + \frac{1}{z^n} = 2 \cos(n\theta)$

2nd part done previously in Endsem (Q 2c)

$$\begin{aligned}
 (4) (a) \quad \nabla \left(\frac{1}{r} \right) &= -r^d \hat{r} \\
 \Rightarrow \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \hat{r} &= -r^d \hat{r} \\
 \Rightarrow -\frac{1}{r^2} \hat{r} &= -r^d \hat{r} \Rightarrow -r^{d+1} \hat{r} \quad [\hat{r} = r \hat{r}] \\
 \text{Comparing both sides ;} \quad &\boxed{d = -3} \quad (\text{Ans})
 \end{aligned}$$

(b) For a radial vector function $\vec{f}(r) = f(r) \hat{r}$:

$\nabla \cdot \vec{f}(r) = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 f)$

Using this:

$$\begin{aligned}
 \text{LHS} &= \nabla \cdot (r^d \hat{r}) \\
 &= \nabla \cdot (r^2 \hat{r}) \\
 &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^4) \\
 &= 4r = dr = \text{RHS}
 \end{aligned}$$

$\therefore \boxed{d = 4} \quad (\text{Ans})$

(c) $\nabla \cdot \nabla (r^3)$

$$\begin{aligned}
 &= \nabla \cdot \left[\frac{\partial}{\partial r} (r^3) \hat{r} \right] = \nabla \cdot (3r^2 \hat{r}) \\
 &= \frac{1}{r^2} \frac{\partial}{\partial r} [r^2 \cdot 3r^2] \\
 &= \frac{1}{r^2} \cdot 12r^3 = 12r = dr
 \end{aligned}$$

$\therefore \boxed{d = 12} \quad (\text{Ans})$

(d) $\left[\begin{array}{l} \text{Curl of gradient} = 0 \\ \text{Divergence of curl} = 0 \end{array} \right]$ Hence, this identity is true for any d .

(5) (a) Done previously in HW-3 (Q 4b) (27)

(b) Let us first prove that $\det(AB) = \det(A) \det(B)$.
Once this is proved, we can assume $X = AB$ & $Y = CD$ to show that

$$\det(XY) = \det(X) \det(Y)$$

$$\Rightarrow \det(ABCD) = \det(AB) \det(CD) \\ = \det(A) \det(B) \det(C) \det(D)$$

$\therefore \det(A) = \pm \epsilon_{i_1 i_2 i_3 \dots i_n}$ in $A_{1i_1} A_{2i_2} A_{3i_3} \dots A_{ni_n}$
for an $n \times n$ matrix A

$$\therefore \det(AB) = \pm \epsilon_{i_1 i_2 i_3 \dots i_n} \text{ in } (AB)_{1i_1} (AB)_{2i_2} \dots (AB)_{ni_n}$$

$$= \epsilon_{i_1 i_2 \dots i_n} \text{ in } (A_{1k_1} B_{k_1 i_1}) (A_{2k_2} B_{k_2 i_2}) \dots (A_{nk_n} B_{k_n i_n})$$

$$= (\epsilon_{i_1 i_2 \dots i_n} \text{ in } B_{k_1 i_1} B_{k_2 i_2} \dots B_{k_n i_n}) (A_{1k_1} A_{2k_2} \dots A_{nk_n})$$

$$= \det(B) \epsilon_{k_1 k_2 \dots k_n} (A_{1k_1} A_{2k_2} \dots A_{nk_n})$$

[using the identity

$$\epsilon_{i_1 i_2 \dots i_n} \text{ in } B_{k_1 i_1} B_{k_2 i_2} \dots B_{k_n i_n}$$

$$= \det(B) \epsilon_{k_1 k_2 \dots k_n}]$$

$$\boxed{\det(AB) = \det(B) \det(A)} \quad \text{(Proved)}$$

Assuming $X = AB$ & $Y = CD$,

$$\det(ABCD) = \det(A) \det(B) \det(C) \det(D)$$

$$\Rightarrow \boxed{\det(ABCD) = \det(DABC)} \quad [\because \text{multiplication is commutative}]$$

(Proved)

(3) (a) The Taylor series expansion of a function is

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n \quad \text{at } z=0$$

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around $z=0$.

let us calculate $f^{(n)}(z)$, $f^{(n)}(0)$

$$\begin{aligned} f^{(n)}(0) &= \left. \frac{d^n}{dz^n} \sum_{k=0}^{\infty} (-1)^{k+1} \sin\left(\frac{z}{k}\right) \right|_{z=0} \\ &= \sum_{k=1}^{\infty} (-1)^{k+1} \left[\frac{d^n}{dz^n} \sin\left(\frac{z}{k}\right) \right] \Big|_{z=0} \end{aligned}$$

① = 0 if n is even, as it gives

$$\sin\left(\frac{z}{k}\right) = 0 \quad \text{at } z=0.$$

If n is odd; it gives $\cos\left(\frac{z}{k}\right) = 1$ at $z=0$.

$$f^{(n)}(0) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^n} (-1)^{\frac{n-1}{2}} \quad ; \quad n = \text{odd}$$

$[(-1)^{\frac{n-1}{2}}$ comes as a result of differentiation of $\cos\left(\frac{z}{k}\right)$ giving $(-1) \sin\left(\frac{z}{k}\right)$]

$$f(z) = \sum_{\substack{n=1 \text{ to } \infty \\ (\text{odd})}} \frac{(-1)^{k+1}}{k^n} (-1)^{\frac{n-1}{2}} z^n \quad \text{Ans}$$

For a series of the form

$$f(z) = \sum_{n=0}^{\infty} a_n z^n ;$$

the radius of convergence can be written as

$$\frac{1}{R} = \lim_{n \rightarrow \infty} |a_n|^{1/n}$$

The above series can be written as:

$$f(z) = \sum_{\substack{n=1 \text{ to } \infty \\ (\text{odd})}} \frac{(-1)^{\frac{n-1}{2}}}{n!} \left[\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^n} \right] z^n$$

$$\therefore a_n = \frac{(-1)^{\frac{n-1}{2}}}{n!} \left[\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^n} \right] = \frac{(-1)^{\frac{n-1}{2}}}{n!} \eta(n)$$

$$\therefore \frac{1}{R} = \lim_{n \rightarrow \infty} |a_n|^{1/n}$$

$$= \lim_{n \rightarrow \infty} \frac{|\eta(n)|^{1/n}}{(n!)^{1/n}}$$

$$\eta(n) = \frac{1}{1!} - \frac{1}{2^n} + \frac{1}{3^n} - \frac{1}{4^n} + \dots$$

$$@ \quad n \rightarrow \infty ; \quad \eta(n) \rightarrow 1 ; \quad [\eta(n)]^{1/n} \rightarrow 1$$

$$@ \quad n \rightarrow \infty ; \quad (n!)^{1/n} \rightarrow \infty$$

$$\therefore \lim_{n \rightarrow \infty} |a_n|^{1/n} = \frac{1}{\infty} \rightarrow 0$$

$$\therefore \frac{1}{R} \rightarrow 0 \Rightarrow \boxed{R \rightarrow \infty} \quad (\text{Ans})$$

The series converges for all finite values of z .

(b) (1) Done previously in Endsem (Q1a)

$$(ii) \quad f(z) = \sinh\left(\frac{1}{z}\right)$$

$$\sinh(u) = u + \frac{u^3}{3!} + \frac{u^5}{5!} + \dots$$

$$\text{Putting } u = \frac{1}{z} :$$

$$\sinh\left(\frac{1}{z}\right) = \frac{1}{z} + \frac{1}{3!z^3} + \frac{1}{5!z^5} + \dots$$

It contains infinitely many powers of z .

Hence, $\boxed{z = 0 \rightarrow \text{essential singularity}} \quad (\text{Ans})$

$$@ \quad z \rightarrow \infty ; \quad u \rightarrow 0$$

$$\therefore \sinh(0) = 0$$

Hence, $\boxed{z = \infty \text{ is a removable singularity}} \quad (\text{Ans})$